# Creating Apps on Live Cloud Platforms: IBM Bluemix and Red Hat OpenShift

Cloud platforms are becoming rather popular with developers for various well known reasons. Through the use of virtual machines, developers can avail of the various cloud services providers offer. This article serves as a primer on creating apps on IBM Bluemix and Red Hat OpenShift.

Nowadays, most computing services that are hosted by a number of cloud service providers are available on demand. Currently, a number of key areas in cloud computing are being researched due to the escalating needs in the digital services domain.

## Virtualisation and hypervisors

Virtualisation is the key technology that works at the back end of cloud services and digital infrastructure. In actual implementation, the cloud is deployed using virtualisation. Whenever there is a need for a remote system by any user, dynamically created virtual machines are provided to the end user or developer at the other remote location with the credentials of the machine at the cloud data centre.

A virtual machine (VM) refers to the software implementation of any computing device, machine or computer which executes the instructions or programs exactly as the physical machine would. When a user or developer works on a virtual machine, the resources, including all programs installed on the remote machine, are accessible using a specific set of protocols. Here, for the end user of the cloud service, the virtual machine acts like the actual machine.

Widely used virtualisation software includes VMware Workstation, VMlite Workstation, Virtual Box, Hyper-V, Microsoft Virtual PC, Xen and KVM (Kernel Based Virtual Machine).

## Hypervisor technology

A hypervisor or virtual machine monitor (VMM) is a piece of computer software, firmware or hardware that creates and runs virtual machines. A computer on which a hypervisor is running one or more virtual machines is defined as a host machine. Each virtual machine is called a guest machine. The hypervisor presents the guest operating systems with a virtual operating platform and manages the execution of these systems. Multiple instances of a variety of operating systems may share the virtualised hardware resources.